Provided that $\sum_{i=1}^{n} p_i = 1; p_i \geq 0$

If E(x)=m, the mathematical expectation of the variable$(x - m)^2$isknownas the variance of the variable x, which is denoted by Var(x).

$$Var(x) = E(x-m)^2 = (x_1 - m)^2 p_1 + (x_2 - m)^2 p_2 + (x_3 - m)^2 p_3 + ... +$$

$$(x_n - m)^2 p_n = \sum_{i=1}^{n} (x_i - m)^2 p_i.$$

We can show that $Var(x) = E(x - m)^2 = E(x^2) - [E(x)]^2$. Remember, in our notation, E(x) = m. The square root of Var(x) is called the standard deviation of the variable x.

If g(x) is any function of the variable x, defined for every value of the variable x, then the expected value of the function (x), denoted by E[g(x)] is given by

$$\sum g(x_i)p_i = g(x_1)p_1 + g(x_2)p_2 + g(x_3)p_3 + ... + g(x_n)p_n = \sum_{i=1}^{n} (X_i - m)^2 p_i$$

Mathematical expectation of a variable is analogous to weighted Arithmetic Mean of the variable, say, x. In a frequency distribution, the relative frequencies (Class frequency / Total frequency) of a variable could be thought of as the probability that the variable will take that value, i.e. $p_i=f_i/N$,

where p_i : Probability that the variable x will take the value x_i

f_i : frequency of occurrence of x_i

N: Total frequency $\sum_{i=1}^{n} f_i$

As we know weighted arithmetic mean $= \dfrac{\sum_{i=1}^{n} X_i f_i}{\sum_{i=1}^{n} f_i} = \dfrac{\sum_{i=1}^{n} X_i f_i}{N} = \sum_{i=1}^{n} X_i p_i = E(x)$

= Expected value of the variable x.

Example: What is the mathematical expectation of the number of points when a nun biased die is thrown?

Let the variable denote the number of points when a die is thrown. Therefore, *x*can take values 1, 2, 3, 4, 5 and 6. The probability of realization of all these values is same, viz.,1/6,therefore,

E(*x*)=1/6(1+2+3+4+5+6)=3.5

Theorem 1

The mathematical expectation of sum of several random variables is equal to the sum of the mathematical expectation of the random variables.

E(a+b+c+d+.........) = E(a)+E(b)+E(c)+E(d)+......., where a, b, c, d represent random variables.

We will consider the theorem for two random variables and the result could be extended to any number of random variables.

Let, x and y are two random variables; x can take the values $x_1, x_2, x_3, \ldots, x_n$ and y can take the values $y_1, y_2, y_3, \ldots, y_m$, and p_{ij} is the probability of the event that $x = x_i$ and $y = y_j$. Now $(x + y)$ is a new random variable and it takes the value $(x_i + y_j)$ with probability p_{ij} or, $P(x = x_i \text{ and } y = y_j) = p_{ij}$. Using the definition of mathematical expectation

$E(x+y) = \sum_{i=1}^{n} \sum_{j=1}^{m} (x_i + y_i) p_{ij}$ [we take double summation as i can take n values and j can take m values]

$$E(x+y) = \sum_{i=1}^{n} \sum_{j=1}^{m} (x_i p_{ij} + y_j p_{ij}) = \sum_{i=1}^{n} \sum_{j=1}^{m} x_i p_{ij} + \sum_{i=1}^{n} \sum_{i=1}^{m} y_j p_{ij}$$

$= \sum_{i=1}^{n} x_i \sum_{j=1}^{m} p_{ij} + \sum_{j=1}^{m} y_j \sum_{i=1}^{n} p_{ij}$ [we could write this as x_i is constant with respect to variations in j]

The following table will explain how we could write the above. In this context, we define marginal probability of a variable given the other variable takes some specific value. Suppose n= 9 and m= 8, i.e., the variables x and y take 9 and 8 values respectively. From the table, it is easy to see the distribution of the variables and their probabilities, where the symbols have their usual meanings.

Marginal probability of y takes the value $yj = \sum_{i=1}^{9} p_{i0} = \sum_{j=1}^{8} p_{oj}$

X ↓ \ y →	y1	y2	y3	y4	y5	y6	y7	y8	Marginal Probability of x ↓
x1	p11	p12	p13	p14	p15	p16	p17	p18	p10
x2	p21	p22	p23	p24	p25	p26	p27	p28	p20
x3	p31	p32	p33	p34	p35	p36	p37	p38	p30
x4	p41	p42	p43	p44	p45	p46	p47	p48	p40
x5	p51	p52	p53	p54	p55	p56	p57	p58	p50
x6	p61	p62	p63	p64	p65	p66	p67	p68	p60
x7	p71	p72	p73	p74	p75	p76	p77	p78	p70
x8	p81	p82	p83	p84	p85	p86	p87	p88	p80
x9	p91	p92	p93	p94	p95	p96	p97	p98	p90
Marginal Probability of y →	p01	p02	p03	p04	p05	p06	p07	p08	1

Similarly, marginal probability of x takes the value $x_i = \sum_{j=1}^{8} p_{ij} = p_{i0}$

In the above situation, $E(x) = \sum_{i=1}^{9} x_i p_{0i}$ and $E(y) = \sum_{i=1}^{n} y_i p_{j0}$

Additionally, we can also derive the conditional distribution of the variables from the above table. One example will elucidate it. The conditional distribution of x given $y = y_1$is as follows:

	$y = y_1$	Conditional Probability distribution of x given $y = y_1$
x_1	p_{11}	p11 / p01
x_2	p_{21}	p21 / p01
x_3	p_{31}	p31 / p01
x_4	p_{41}	p41 / p01
x5	p_{51}	p51 / p01
x6	p61	p61 / p01
x7	p71	p71 / p01
x8	p81	p81 / p01
x9	p91	p91 / p01
Total	p01	1

Therefore, $E(x+y) = \sum_{i=1}^{n} x_i \sum_{j=1}^{m} p_{ij} + \sum_{j=1}^{m} y_j \sum_{i=1}^{n} p_{ij}$

$$= \sum_{i=1}^{n} x_i p_{i0} + \sum_{j=1}^{m} y_j p_{0j} = E(x) + E(y)$$

Theorem2

The mathematical expectation of product of several independent random variables is equal to the product of the mathematical expectation of the random variables.

We retain the symbols of the previous theorem and additionally, we assume that the variables x and y are independent, i.e., occurrence of any one of the event has no impact on the occurrence of the other. Let the variable x take the value x_i with probability p_i and the variable y takes the value y_j with probability q_j .Since x and y are independent

$P(x = x_i \text{ and } y = y_j) = p_i \times q_j$.

The theorem states that

$E(x.y) = E(x) \times E(y)$.

Using the definition of mathematical expectation

$$E(x,y)=\sum_{i=1}^{n}\sum_{j=1}^{m} x_i y_j p(x=x_i \text{ and } y=y_j)=\sum_{i=1}^{n}\sum_{j=1}^{m} x_i y_j (p\times q_j)$$

Summing over j and keeping i constant

$$=\sum_{i=1}^{n} x_i p_i \times \sum_{j=1}^{m} y_j q_j)=E(x)\times E(y).$$

This theorem could be extended to any number of variables.

Corollary 1: If m is the expected value of the variable x, then

$$E(x-m)^2=E(x^2)-[E(x)]^2.$$

Proof: $E(x-m)^2=\sum_{i=1}^{n}(x-m)^2 p_i$

$=\sum (x^2-2xm+m^2)p_i$

$=\sum (x^2)p_i- 2m\sum xp_i+m^2\sum p_i$

$=E(x^2)-2mE(x)+m^2$

$=E(x^2)-2m.m+m^2$

$=E(x^2)-m^2$

$=E(x^2)-(E(x))^2$

$E(x-E(x))^2$ is called the variance of the variable x and it is denoted by Var(x).

Corollary 2: If there are n random variables $x_1,x_2,x_3,\ldots\ldots,x_n$ each having mean $m_1, m_2, m_3,..,m_n$.

$$Var(x_1+x_2+x_3+\ldots+x_n)=\sum_{i=1}^{n} Var(x_i)+2\sum_{i=i}^{n} i\neq j\,Cov(x_i,x_j)$$

$Cov(x_i,x_j)=E(x_i-m_i)(x_j-m_j)$ when $\neq j$.

$$Var(x_1+x_2+x_3+\ldots+x_n)=E[(x_1+x_2+x_3+\ldots+x_n)-(m_1+m_2+m_3+\ldots+m_n)]^2$$

$$=\sum_{i=1}^{n} E(x_i-m_i)+2\sum_{i=1}^{n} i\neq j\,E(x_i-m_i)(x_j-m_j)$$

The first term in the above equation is Variance of the variable x_i whereas the second term is the $Cov(x_i,x_j)$. Therefore,

$$Var(x_1+x_2+x_3+\ldots+x_n)=\sum_{i=1}^{n} Var(x_i)+2\sum_{i=1}^{n} i\neq j\,Cov(x_i,x_j)$$

If the variables are mutually independent then the covariance term in the above expression is zero (since mutual independence rules out joint occurrence).

Check Your Progress 5

1) A man purchases a lottery ticket. He may win first prize of Rs.10,000 with probability .0001 or the second prize of Rs. 4,000 with probability .0004. On an average, how much he can expect from the lottery?

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2) A box contains 4 white balls and 6 black balls. If 3 balls are drawn at random, find the mathematical expectation of the number of white balls.

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3) If y=a + b.x, where a and b are constants, prove that E(y)=a +bE(x).

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25.9 LET US SUM UP

The term probability in a very crude sense implies the chance factor and it issued frequently where there is uncertainty about something. Mathematicians from the very early ages tried endlessly to build up a structure under which this uncertain thing called probability could be analysed. In this unit we learnt about some basic terminology which is used in probability related matters. We discussed various methods frequently used to determine the probability of various non-deterministic experiments. Using these techniques, you can determine the probabilities of many day-to-day uncertain events. Further we learnt about the theorems of probability (total and compound) and conditional probability. Mathematical expectation introduced at the end of this unit is nothing but the mean of a random variable. It is quite useful in understanding the nature of a random variable.

25.10 KEY WORDS

Bayes' Theorem : If an event A occurs in conjunction with one of them mutually exclusive and collectively exhaustive events E_1, E_2, E_3..... E_n and if A actually occurs, then the probability that it was preceded by a particular event E_i(I=1,2,3……n)is given by

$$P(B_i \mid A) = P(A \mid B_i) \times P(A) / \sum_{i=1}^{n} P(B_i)P(A \mid B_i)$$

Conditional Events : If two events A and B are mutually exclusive and if it is known that event b has already taken place, the probability of A is known as the conditional probability of A given B. Symbolically P(A|B) = P (A ∩ B) / P(B).

Events : When some of the elements of the sample space of a random experiment satisfy a particular criterion, we call it an event.

Independent Events : Events are said to be independent of each other if occurrence of one event is not affected by the occurrence of the others. If events A and B are independent then P(A∩B)=P(A)×P(B).

Marginal Probability : In a bivariate distribution the probability that X will assume a given value X whatever the value of Y is called the marginal probability of Y. Similarly, one c and e fine the marginal distribution of Y.

Mathematical Expectation : If the variable x can take values x_1, x_2, x_3,.........,x_n with probabilities p_1, p_2, p_3,.........,p_n, then the mathematical expectation or the expected value of the variable, is defined as the weighted sum of the values of the variable the weights being the probabilities of the values of the variable. Expected value of a variable 'x' is denoted by E(x).

$$E(x) = x_i p_1 + x_2 p_2 + x_3 p_3 + ... + x_n p_n = \sum_{i=1}^{n} x_i p_i$$

Provided that $\sum_{i=1}^{n} p_i = 1$.

Mutually Exclusive Events : Events are said to be mutually exclusive when no two or more of them can occur simultaneously.

Mutually Exhaustive Events : Events are exhaustive if at least one of them necessarily occurs.

Sample Space : The collection of all possible outcomes of an experiment is called the sample space.

25.11 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) If the three children were born on three different days, the first child may be born in any one of 365 days, the second children has to be born on anyone of the remaining 364 days and the third child has to be born on anyone of the remaining 363 days. Therefore, the probability that the three children will be born on 3 different days in a year is 365.364.363/365.365.365.

2) Five persons can sit in a row in 5! = 5.4.3.2.1 = 120 ways. Considering a and b together they can arrange among themselves in 4! 2!= 48 ways. Therefore, the required probability is = 48/120 = .4.

3) Two cards can be drawn from a pack of 52 cards in ${}^{52}C_2$= 1326 ways. These outcomes are mutually exclusive, exhaustive and equally likely.

   a) The number of cases favourable to both the cards are red is ${}^{26}C_2$= 325. Therefore, that probability of drawing both red cards = 325/1326.

   b) One heart and one diamond can be drawn in 13.13= 169 ways. Therefore, probability of drawing one heart and one diamond= 169/1326.

4) One white ball could be drawnin${}^{6}C_1$=6 ways and one ball can be drawn in ${}^{10}C_1$ = 10 ways. Therefore, the probability of drawing one white ball =6/10=.6.

5) The total number of ways of distributing n identical objects into r compartments is given by the formula ${}^{n+r-1}C_{r-1}$. Using the formula we can find out that there are 816 mutually exclusive exhaustive and equally likely ways of distributing 15 identical objects into 4 numbered boxes.

   a) If each box is to contain at least 2 objects, we place 2 objects in each box and then the remaining 7 objects could be distributed among the 4 boxes. Using the above formula, we get there are 120 ways of doing that. Therefore, the required probability is 5/34.

   b) If the number box has to be empty then it means there should be at least one object in each box. We first distribute 1 object in each box. Then the remaining 11 objects could be distributed in 364 ways (using the above-mentioned formula). Therefore, the required probability is given by 91/204.

6) Total number of possible equally likely exhaustive and exclusive ways of choosing 3 tickets is given by ${}^{20}C_3$ = 1140. The three numbers will be in A.P if the difference between the numbers is either 1 or 2 or 3 ……..9. If the

difference is 1 then 18 sets of numbers are possible, (123,234,345,………181920). Similarly, if the difference is 2 then 16 sets of numbers are possible, (135,246,357,………161820).Thus, we can find the total number of sets of 3numbers all of them being in A. P is 90. and the required probability is 3/38.

Check Your Progress 2

1) $P(A^c)=½, P(AUB)=7/12, P(A|B)=3/4, P(A^c \cap B)=1/12, P(A^c \cap B^c) = 5/12$, $P(A^c U\ B) = 3/4$.

2) Do yourself using Sub-section 25.5.1.1.

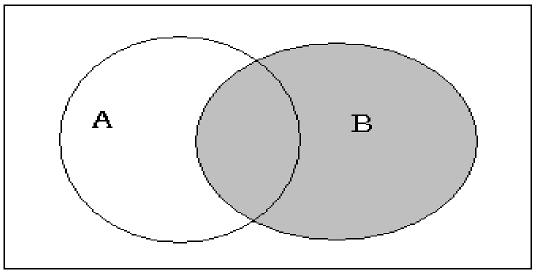
Check Your Progress 3

1) A: The sales person makes sale to the first customer.

B: The sales person makes sale to the second customer.

P(A)= P(B)= ½

(AU B): the sales person will make a sell.

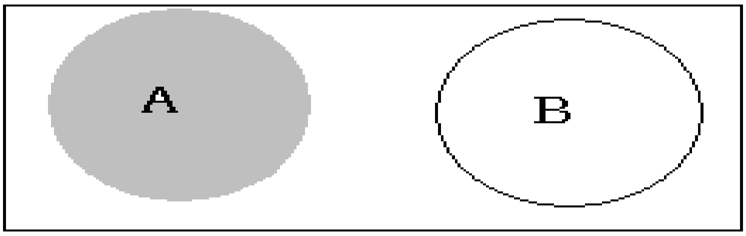


$P(AUB)=1-P(A^c \cap B^c)=1-P(A^c)(B^c)$[as events A and B are independent]

$=1–½. ½=¾$

2) Do yourself using Sub-section 25.6.2.

3)



$P(A\,|\,A \cup B) = P(A \cap (A \cup B)) / P(A) + P(B).[P(A \cup B) = P(A) + P(B)$ theorem of total probability]

$= P((A \cap A) \cup (A \cap B)) = P(A \cup (A \cap B)) = P(A) + P(A \cap B) - P(A \cap (A \cap B))$

$= P(A) + 0 - 0 = P(A)$

Therefore $P(A\,|\,A \cup B) = P(A) / P(A) + P(B)$

4) Hints: Let A denote the event of getting 3 white balls in the first draw and B denote the events of getting 3 red balls in the second draw. We need to find the compound probability P(AB) = P(A) P(B|A)

Check Your Progress 4

1) A: Selected bulb is defective.

B_1: Selected bulb has been produced by machine 'a'

B_2: Selected bulb has been produced by machine 'b'

B_3: Selected bulb has been produced by machine 'c'

Bi	**P(Bi)**	**P(A|Bi)**	**P(Bi).P(A|Bi)**
B1	0.25	0.05	0.0125
B2	0.35	0.04	0.014
B3	0.4	0.02	0.008
Total	1		0.0345

From Bayes' theorem

$$P(B_3 \mid A) = P(A \mid B_3) \times P(A) / \sum_{i=1}^{n} P(B_i)\, P(A \mid B_i) = .008 / 0.345 = 16/69.$$

2) Follow the same method. Answer=3/7.

Check Your Progress 5

1) The calculations for mathematical expectation are provided below

X	P(x)	x.P(x)
0	0.9995	0
4000	0.0004	1.6
10000	0.0001	1
Total		2.6

E(x)= 2.6Rs.

2) Probability of drawing '0' white ball = ${}^4C_0\,{}^6C_3/{}^{10}C_3$ = 1/6

Probability of drawing '1' white ball = ${}^4C_1\,{}^6C_2/{}^{10}C_3$ = ½

Probability of drawing '2' white ball = ${}^4C_2\,{}^6C_1/{}^{10}C_3$ = 3/10

Probability of drawing '3'white ball=${}^4C_3\,{}^6C_0/{}^{10}C_3$= 1/30

Fill up the following blank table to get the required expectation

X	p(x)	x.P(x)
0 1 2 3		
Total		

3) Let x takes the values $x_1, x_2, ..., x_n$ with probabilities $p_1, p_2, ... p_n$ respectively, where $\sum p_i = 1$. Therefore, y will take the values:
$y_1 = ax_1 + b, y_2 = ax_2 + b, ..., y_n = ax_n + b$

$$E(y) = y_1 p_1 + y_2 p_2 + y_n p_n = (ax_1 + b)p_1 + (ax_2 + b)p_2 + ... + (ax_n + b)p_n$$
$$E(y) = (ax_1 p_1 + ax_2 p_2 + ... + ax_n p_n) + (bp_1 + bp_2 + ... + bp_n)$$

$$E(y) = \sum_{i=1}^{n} ax_i p_i + \sum_{i=1}^{n} bp_i = a\sum_{i=1}^{n} x_i p_i + \sum_{i=1}^{n} bp_i$$

25.12 EXERCISES

Q1. There are 5 green and 7 red balls. Two balls are selected one by one without replacement. Find the probability that first is green and second is red.

Ans. P(G) X P(R) = (5/12) X (7/11) = 35/132

Q2. Two cards are drawn from the pack of 52 cards. Find the probability that both are diamond or both are kings.

Ans. Total number of ways = ${}^{52}C_2$

Both are diamonds = ${}^{13}C_2$

Both are Kings = ${}^{4}C_2$

P (both are diamonds or both are kings) = $({}^{13}C_2 + {}^{4}C_2) / {}^{52}C_2$

= 0.063

Q3. A tyre manufacturing company kept a record of the distance covered before a tyre needed to be replaced. The table shows the results of 1000 cases

Distance (in Km)	Less than 4000	4001 to 9000	9001 to 14000	More than 14000
Frequency	20	210	325	445

If a tyre is bought from this company, what is the probability that:

i) it has to be substituted before 4000 km is covered?

ii) it will last more than 9000 km?

iii) it has to be replaced between 4000 km and 14000 km covered by it?